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DESIGN DOCUMENT

Project Expenses Tracker

Thailand Fieldtrip Edition

thai.expenses.syaefulaz.my.id

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1. Document Purpose

This Design Document (DD) provides the technical architecture and implementation specification for the Thailand Expenses Tracker system. It complements BRD-TXC-EXP-2025-001 by detailing how the requirements will be implemented.

2. System Overview

2.1 Architecture

Layer	Technology	Notes
Presentation	Angular 21 + PrimeNG	SPA served by Nginx
Application	NodeJS 18 + Express	REST API, TypeScript
Data	MySQL 8	InnoDB, transactions
Web Server	Nginx 1.24+	Reverse proxy, SSL termination
Authentication	JWT	Access (15-min) + Refresh (7-day)

2.2 Data Flow

Step	Actor / Component	Action
1	User	Opens browser → https://thai.expenses.syaefulaz.my.id
2	Nginx	Proxies /api/* → NodeJS; serves Angular static
3	Angular SPA	User fills expense form (date, amount, source, description)
4	NodeJS API	If source ≠ recorder → auto-creates loan record
5	MySQL	Stores expense + loan atomically; returns confirmation
6	Dashboard	Shows live loan balances: who owes whom how much

2.3 Technology Stack

Component	Technology	Version
UI	Angular	21.x
Component Library	PrimeNG	18.x
Backend	NodeJS	18.x LTS
API	Express	—
ORM	TypeORM	Latest
Database	MySQL	8.x
Web Server	Nginx	1.24+
SSL	Let's Encrypt (Certbot)	—
Auth	JWT + bcrypt	—

3. Frontend Design

3.1 Page Structure (5 Tabs)

Tab	Route	Description
Dashboard	/dashboard	Net position summary; who owes whom; quick KPIs
List of Data	/expenses	Expense table with filter/sort; New Input button
Loan Data	/loans	All loans; declare repayment; record actual payment
History	/history	Submitted / approved / rejected expenses audit log
Admin	/admin	User management (ADMIN/Syaeful only)

3.2 Navigation

Top navigation bar with 5 tabs: Dashboard | List of Data | Loan Data | History | Admin. A persistent loan badge shows total outstanding balance for the logged-in user.

3.3 Expense Entry Form (New Input)

Fields for new expense record:

Field	Type	Required	Notes
expense_date	Calendar	Yes	Date expense occurred — NOT submission date
amount	InputNumber	Yes	Numeric; 2 decimal places
description	InputText	Yes	Free-text description
source	Dropdown	Yes	Winda Cash, ATM Winda, Syaeful Cash, Dina Cash, ATM Dina, Corporate
category	Dropdown	No	Transport, Food, Accommodation, Other
attachment	FileUpload	No	Receipt image/PDF

3.4 Loan Dashboard Flow

Two views within the Loan Data tab:

View	Description
Money I Lent	List of unsettled loans where logged-in user is lender. Shows borrower name, original amount (THB), declared repayment, actual repaid, remaining balance, status. Actions: Declare Repayment Amount, Record Actual Repayment (partial/full).
Money I Borrowed	List of loans where logged-in user is borrower. Shows lender, amount, remaining balance. Read-only.

4. Backend Design

4.1 Key API Endpoints

Method	Endpoint	Description
POST	/api/v1/auth/login	Login; returns JWT access + refresh tokens
POST	/api/v1/auth/refresh	Refresh access token
GET	/api/v1/expenses	List expenses (filter by date, user, source)
POST	/api/v1/expenses	Create expense; if source ≠ user → auto-create loan
PUT	/api/v1/expenses/:id	Update draft expense
DELETE	/api/v1/expenses/:id	Delete draft expense
POST	/api/v1/expenses/:id/submit	Submit for approval → status PENDING
GET	/api/v1/loans	List all loans for current user (as lender + borrower)
PUT	/api/v1/loans/:id/declare	Lender declares expected repayment amount
PUT	/api/v1/loans/:id/repay	Lender records actual repayment; updates balance
GET	/api/v1/loans/summary	Net position: total lent, total borrowed, net balance
GET	/api/v1/dashboard	KPIs: total expenses, loan summary, recent activity
GET	/api/v1/history	Audit log: all submitted/approved/rejected expenses
GET	/api/v1/users	List users (ADMIN only)
POST	/api/v1/users	Create user (ADMIN only)
PUT	/api/v1/users/:id	Update user role/status (ADMIN only)

4.2 Expense Request Schema (POST /api/v1/expenses)

Field	Type	Required	Notes
expense_date	DATE	Yes	Date expense occurred (YYYY-MM-DD)
amount	DECIMAL(15,2)	Yes	Amount in THB
description	TEXT	Yes	Description of expense
source	VARCHAR(100)	Yes	Who funded: Winda Cash, ATM Syaeful, etc.
category	VARCHAR(100)	No	e.g. Transport, Food, Accommodation, Other
attachment	FILE	No	Receipt image/PDF

4.3 Loan Auto-Creation Logic

On POST /api/v1/expenses, the API checks if expense.source belongs to a different user. If yes, a loan record is atomically created:

Field	Value
lender_id	ID of the user whose source was used (looked up from source name)
borrower_id	ID of the user creating the expense
amount	expense.amount (original loan in THB)
declared_repayment	NULL (set later by lender)
actual_repaid	0
status	UNSETTLED

5. Database Design (MySQL 8)

5.1 Tables

Core tables:

users

Column	Type	Constraints	Notes
id	INT	PK, AUTO_INCREMENT	
name	VARCHAR(100)	NOT NULL	Display name: Dina, Winda, Syaeful
email	VARCHAR(255)	UNIQUE, NOT NULL	Login email
password_hash	VARCHAR(255)	NOT NULL	bcrypt hashed
role	ENUM('ADMIN','MANAGER')	NOT NULL	ADMIN = Syaeful; MANAGER = Dina/Winda
is_active	BOOLEAN	DEFAULT TRUE	
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	
updated_at	TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP	

expenses

Column	Type	Constraints	Notes
id	INT	PK, AUTO_INCREMENT	

Column	Type	Constraints	Notes
user_id	INT	FK → users.id, NOT NULL	User who created the expense
expense_date	DATE	NOT NULL	Date expense occurred
amount	DECIMAL(15,2)	NOT NULL	Amount in THB
description	TEXT	NOT NULL	Description
source	VARCHAR(100)	NOT NULL	Who funded: Winda Cash, ATM Syaeful, etc.
category	VARCHAR(100)	NULL	Transport, Food, Accommodation, Other
attachment_path	VARCHAR(500)	NULL	Path to receipt file
status	ENUM('DRAFT','PENDING','APPROVED','REJECTED')	DEFAULT DRAFT	Workflow status
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	
updated_at	TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP	

loans [NEW]

Column	Type	Constraints	Notes
id	INT	PK, AUTO_INCREMENT	
expense_id	INT	FK → expenses.id, UNIQUE	1:1 link to source expense
lender_id	INT	FK → users.id, NOT NULL	User whose source funds were used
borrower_id	INT	FK → users.id, NOT NULL	User who recorded the expense
amount	DECIMAL(15,2)	NOT NULL	Original loan amount (THB)
declared_repayment	DECIMAL(15,2)	NULL	Amount borrower says they will repay
actual_repaid	DECIMAL(15,2)	DEFAULT 0	Amount actually repaid so far

Column	Type	Constraints	Notes
remaining_balance	DECIMAL(15,2)	Computed	declared_repayment – actual_repaid
status	ENUM('UNSETTLED','PARTIAL','FULLY_SETTLED','OVERPAID')	DEFAULT UNSETTLED	Lifecycle state
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	
updated_at	TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP	

loan_repayments [NEW]

Column	Type	Constraints	Notes
id	INT	PK, AUTO_INCREMENT	
loan_id	INT	FK → loans.id, NOT NULL	
recorded_by_id	INT	FK → users.id, NOT NULL	Lender who recorded
amount	DECIMAL(15,2)	NOT NULL	Repayment increment (THB)
note	TEXT	NULL	Optional note
repaid_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	

6. Security Design

Area	Implementation
HTTPS	SSL/TLS enforced on all traffic
Passwords	bcrypt cost 12; never stored in plaintext
Auth	JWT access (15-min) + refresh (7-day)
RBAC	ADMIN → full; MANAGER → expense + loan CRUD; no ADMIN tab for MANAGER
Input Validation	Express-validator on all API inputs
SQL Injection	Parameterized queries via TypeORM
CORS	Nginx CORS headers restricting to known origins
Rate Limiting	Nginx limit_req 100 req/min per IP on /api/

7. Infrastructure & Deployment

Item	Detail
Domain	thai.expenses.syaefulaz.my.id
Deploy Path	/opt/expenses-tracker/
Server	Texcoms VPS
OS	Ubuntu 22.04 LTS
Nginx	Reverse proxy on port 443 (SSL), port 80 → redirect
NodeJS API	PM2 process manager; internal port 3000
Angular Build	Static files served by Nginx at /
MySQL	Port 3306, localhost only
SSL	Let's Encrypt via Certbot, auto-renewal cron

7.1 Nginx Config Snippet

```
server {  
  
    listen 80;  
  
    server_name thai.expenses.syaefulaz.my.id;  
  
    return 301 https://$host$request_uri;  
}  
  
server {  
  
    listen 443 ssl http2;  
  
    server_name thai.expenses.syaefulaz.my.id;  
  
    ssl_certificate /etc/letsencrypt/.../fullchain.pem;  
    ssl_certificate_key /etc/letsencrypt/.../privkey.pem;  
  
    location / {  
  
        root /opt/expenses-tracker/dist;  
        try_files $uri $uri/ /index.html;  
    }  
  
    location /api/ {  
  
        proxy_pass http://127.0.0.1:3000;  
    }  
}
```

8. Design Acceptance Criteria

#	Acceptance Criterion	Verification
DAC-01	Expense form submits with: date, amount, description, source (all mandatory), category (opt), attachment (opt)	Create expense as MANAGER → expense saved with status DRAFT
DAC-02	Expense with source ≠ recorder auto-creates loan visible on lender's Loan Data tab	Syaeful submits expense with source=Winda Cash → check Loan Data as Winda
DAC-03	Lender can declare repayment amount; remaining balance updates	PUT /api/v1/loans/:id/declare → GET loan → verify remaining_balance
DAC-04	Lender can record partial repayment; loan_repayment audit record created	PUT /api/v1/loans/:id/repay → GET /api/v1/loans/:id → verify repayment log
DAC-05	Dashboard shows accurate net position for each user	Create loans → verify Dashboard figures match sum of loan balances
DAC-06	ADMIN tab visible only to Syaeful; Dina and Winda see no Admin tab	Login as Dina → no Admin link in nav
DAC-07	History tab shows all submitted/approved/rejected expenses per user	Submit expense → check History tab → status shown correctly
DAC-08	HTTPS active at https://thai.expenses.syaefulaz.my.id/	Browser test → green padlock; SSL Labs grade A

End of Document — DD-TXC-EXP-2025-001 v1.0
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